

Week 3: Elementary Functions and Integration (C+B Ch3 and Ch4)

The consequences of having multi-valued functions are unavoidable. To see this we can look at taking powers.

For real numbers: $x^c = e^{c \log x}$

We use the same definition for complex numbers:

$$z^c = e^{c \log z}$$

However, we immediately find that this is a multivalued function of z .

Note if $c \in \mathbb{R}$ then

$$\begin{aligned} z^c &= \exp(c \log z) \\ &= \exp(c(\ln|z| + i(\arg(z) + 2n\pi))) \\ &= \exp(\ln|z|^c + ic(\arg(z) + 2n\pi)) \\ &= |z|^c e^{ic \arg(z)} e^{i2\pi nc} \end{aligned}$$

so we get the normal properties that modulus is raised to the power c , and the argument is multiplied by a power of c .

The additional factors of $e^{i2\pi n}$ then come from the possible non-unique choices of powers.

Question: If $c = \frac{1}{3}$, how do we interpret the values of this function? what if $c = 3$?

However, if c is not real, then things behave a little differently.

Example: What is i^i ?

$$\begin{aligned} i^i &= e^{i \log i} = \exp\{i(\ln|i| + i(\arg(i) + 2\pi n))\} \quad n \in \mathbb{Z} \\ &= \exp\{-\arg(i) - 2\pi n\} \quad n \in \mathbb{Z} \\ &= \exp\left\{-\frac{\pi}{2} + 2\pi n\right\} = e^{(2n - \frac{1}{2})\pi} \quad n \in \mathbb{Z} \end{aligned}$$

These are all real numbers, and they have magnitudes different from 1.

Note that the function $f(z) = z^c$ is analytic on the domain $D = \{ |z| > 0, -\pi < \arg(z) < \pi \}$ because it is a composition of analytic functions. Moreover, by the chain rule:

$$\begin{aligned} \frac{d}{dz} (z^c) &= \frac{d}{dz} (\exp(c \log z)) \\ &= \frac{d}{dz} (\log z) \exp(c \log z) \\ &= \frac{c}{z} \exp(c \log z) \\ &= \frac{c}{\exp(\log z)} \exp(c \log z) \\ &= c \exp((c-1) \log z) = c z^{c-1}. \end{aligned}$$

The principal value of z^c is given by

$$\text{P.v. } z^c = \exp(c \operatorname{Log} z)$$

Example: Let $z_1 = 1+i$, $z_2 = 1-i$, $z_3 = -1-i$

$$\text{Then } (z_1 z_2)^i \stackrel{?}{=} z_1^i z_2^i, \quad (z_2 z_3)^i \stackrel{?}{=} z_2^i z_3^i \quad ?$$

$$z_1 = \sqrt{2} e^{i\pi/4}, \quad z_2 = \sqrt{2} e^{-i\pi/4}$$

$$\Rightarrow z_1 z_2 = 2$$

$$\text{and } (z_1 z_2)^i = 2^i = e^{i \ln 2}$$

$$\text{while } z_1^i = \exp(i(\ln \sqrt{2} + i\frac{\pi}{4}))$$

$$= e^{\pi/4} e^{i\frac{1}{2} \ln 2}$$

$$\text{and } z_2^i = e^{-\pi/4} e^{i\frac{1}{2} \ln 2}$$

$$\Rightarrow z_1^i z_2^i = e^{i \ln 2} \quad \checkmark$$

$$\text{Hence, } (z_1 z_2)^i = z_1^i z_2^i.$$

On the otherhand:

$$z_2 = \sqrt{2} e^{-i\pi/4} \quad \text{and} \quad z_3 = \sqrt{2} e^{-3\pi/4 i}$$

$$\Rightarrow z_2 z_3 = 2 e^{-\pi i}$$

$$\text{while} \quad z_2^i = e^{-\pi/4} e^{i\frac{1}{2}\ln 2}$$

$$\text{and} \quad (z_2 z_3)^i = \exp(i(\ln 2 - \pi i)) \\ = e^{\pi} e^{i\ln 2}$$

$$z_3^i = e^{-3\pi/4} e^{i\frac{1}{2}\ln 2}$$

$$\Rightarrow z_2^i z_3^i = e^{-\pi} e^{i\ln 2}$$

$$\text{So we find: } (z_2 z_3)^i = z_2^i z_3^i e^{-2\pi}.$$

We can also define the exponential base c :

$$f(z) = c^z = \exp(z \log c).$$

In this case we actually get an entire function by choosing the principal logarithm, since e^z and z are both entire functions.

Moreover, by the chain rule:

$$\begin{aligned} \frac{d}{dz} (c^z) &= \frac{d}{dz} (\exp(z \log c)) \\ &= \log c \cdot \exp(z \log c) \\ &= \log c \cdot c^z. \end{aligned}$$

Trigonometric Functions

One last example is trigonometric function. Again we can use the real functions to find the complex ones.

For real numbers x :

$$e^{ix} = \cos x + i \sin x$$

$$\Rightarrow \cos x = \frac{1}{2} (e^{ix} + e^{-ix})$$

$$\sin x = \frac{1}{2i} (e^{ix} - e^{-ix})$$

So we define:

$$\cos z = \frac{1}{2} (e^{iz} + e^{-iz})$$

$$\sin z = \frac{1}{2i} (e^{iz} - e^{-iz})$$

These functions are entire and we can use the differentiation rules

$$\begin{aligned} \frac{d}{dz} (\cos z) &= \frac{1}{2} (ie^{iz} - i\bar{e}^{iz}) \\ &= \frac{i}{2} (e^{iz} - e^{-iz}) \\ &= -\sin z \end{aligned} \qquad \begin{aligned} \frac{d}{dz} (\sin z) &= \frac{1}{2i} (ie^{iz} + i\bar{e}^{iz}) \\ &= \frac{1}{2} (e^{iz} + e^{-iz}) \\ &= \cos z \end{aligned}$$

We also recover Eulers formula, now for all complex numbers:

$$e^{iz} = \cos z + i \sin z;$$

and familiar properties:

$$\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$$

$$\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$$

$$\sin^2 z_1 + \cos^2 z_1 = 1.$$

$$\sin(z + 2\pi) = \sin z, \quad \cos(z + 2\pi) = \cos z.$$

By plugging in $z = iy$ we recover the hyperbolic trig functions.

$$\begin{aligned} \sin(iy) &= \frac{1}{2i} (e^{i^2 y} - e^{-i^2 y}) & \cos(iy) &= \frac{1}{2} (e^{i^2 y} + e^{-i^2 y}) \\ &= \frac{i}{2} (e^y - e^{-y}) & &= \frac{1}{2} (e^y + e^{-y}) \\ &= i \sinh(y) & &= \cosh(y) \end{aligned}$$

Actually more is true.

$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

$$\cos z = \cos x \cosh y - i \sin x \sinh y$$

From this we see that $\sin z$ and $\cos z$ are not bounded functions for $z \in \mathbb{C}$.

By noting that $|\sin z|^2 = \sin^2 x + \sinh^2 y$

$$\text{and } |\cos z|^2 = \cos^2 x + \sinh^2 y$$

we see that the zeros of sine and cosine are exactly the ones found on the real axis.

Lastly we can define

$$\tan z := \frac{\sin z}{\cos z}$$

$$\cot z := \frac{\cos z}{\sin z}$$

$$\sec z := \frac{1}{\cos z}$$

$$\operatorname{cosec} z := \frac{1}{\sin z}$$

These functions are all analytic except where the denominators are zero and

$$\frac{d}{dz} \tan z = \sec^2 z$$

$$\frac{d}{dz} \cot z = -\operatorname{cosec}^2 z$$

$$\frac{d}{dz} \sec z = \sec z \tan z$$

$$\frac{d}{dz} \operatorname{cosec} z = -\operatorname{cosec} z \cot z$$

Inverse trigonometric functions can also be found, but are multivalued. I will leave finding formulas for these, and determining their analyticity to the homework.

Integrals of Complex Functions

Before generalizing to complex numbers, we should recap what we know about integration of real valued functions.

There are two related processes that we refer to by integration.

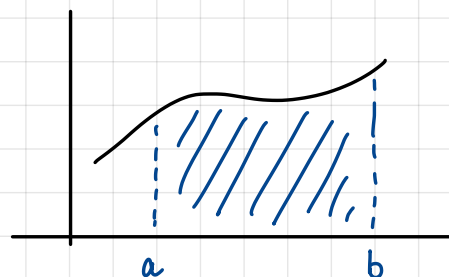
Indefinite integration is the inverse of differentiation.

$$\frac{d}{dx}(f(x)) = g(x) \iff f(x) = \int g(x) dx + C.$$

Definite integration determines the area under a curve:

$$\int_a^b g(x) dx = f(b) - f(a)$$

$$(\text{where } \frac{d}{dx}(f(x)) = g(x)).$$



In order to evaluate a definite integral, it is often useful to use a change of coordinates.

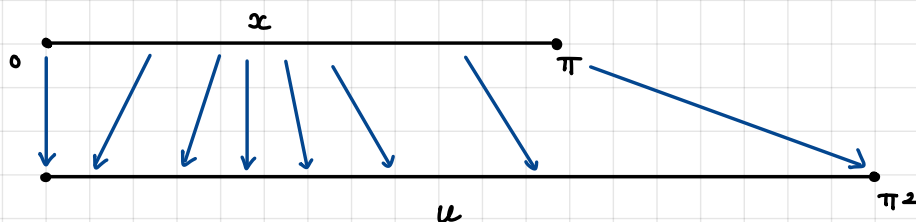
Example: $\int_0^{\pi} 2x \cos(x^2) dx = ?$

$$\text{Let } u = x^2 \Rightarrow du = 2x dx$$

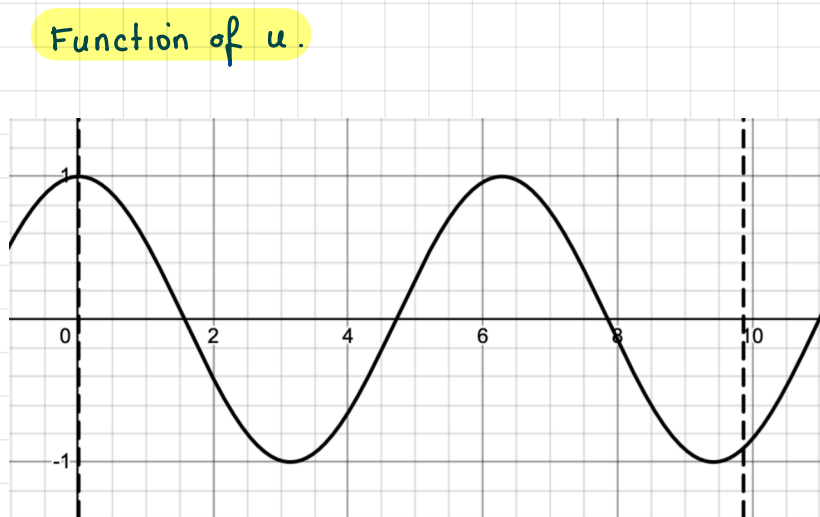
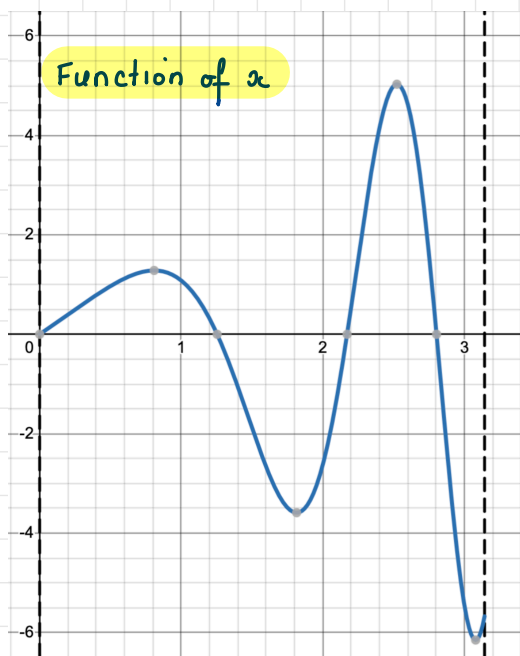
$$x=0 \Rightarrow u=0 \quad \text{and} \quad x=\pi \Rightarrow u=\pi^2$$

$$\therefore \int_0^{\pi} 2x \cos(x^2) dx = \int_0^{\pi^2} \cos(u) du = \sin(u) \Big|_0^{\pi^2} = \sin(\pi^2).$$

But what are we actually doing here?



We are choosing to follow the straight line path from $[0, \pi]$ at a different speed.



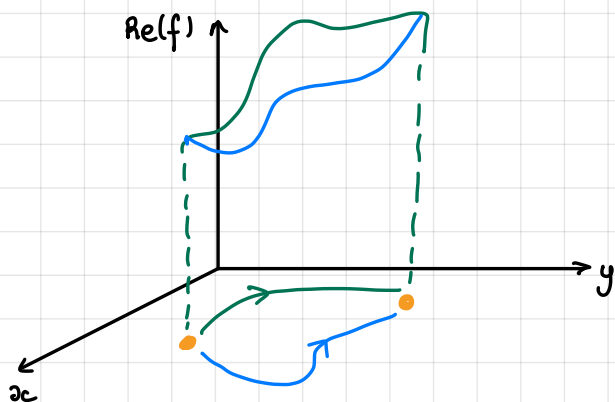
By choosing a particular parameterization of the path $[0, \pi]$, we simplify the integrand to be an elementary function whose anti-derivative is known.

Implicit in this process is the fact that, no matter how we parameterize the path, we get the same value for the integral.

For the complex scenario we want similar things:

- anti-derivatives
- integrals along paths.

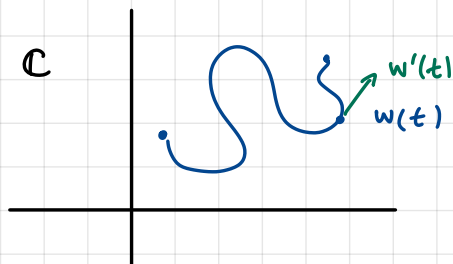
But now things are more complicated because there are many paths between two points.



Should we even expect that these paths would give the same value?

Derivatives of Functions on a Path

Suppose $w: I \subseteq \mathbb{R} \rightarrow \mathbb{C}$ and write $w(t) = u(t) + iv(t)$ for real valued functions u, v . We can think of this as a path through the complex plane.



The derivative of w is

$$w'(t) = u'(t) + iv'(t)$$

can be interpreted as a vector tangent to the path.

We can also start taking complex functions of w to give us new paths.

Example: $\tilde{w}(t) = z_0 w(t)$ for $w_0 \in \mathbb{C}$

Multiplication by the constant z_0 rescales the modulus and shifts the argument of the path w .

We can also differentiate: $\tilde{w}'(t) = z_0 w'(t) = z_0 (u'(t) + iv'(t))$

These paths satisfy many of the rules of calculus for real-valued functions. But, there are some exceptions.

Example: Suppose $w(t)$ is continuous on $a \leq t \leq b$ and differentiable for $a < t < b$.

If w is real then the mean value theorem would say that $\exists c \in (a, b)$ s.t.

$$w'(c) = \frac{w(b) - w(a)}{b - a}$$

For w complex this is no longer the case. E.g. $w(t) = e^{it}$ $t \in (0, 2\pi)$.

Definite Integrals

Let $w(t) = u(t) + iv(t)$ for real-valued u and v and $t \in \mathbb{R}$.

We define:
$$\int_a^b w(t) dt := \int_a^b u(t) dt + i \int_a^b v(t) dt$$

provided the two integrals on the right exist.

We also see that:

$$\operatorname{Re} \left(\int_a^b w(t) dt \right) = \int_a^b \operatorname{Re}(w(t)) dt$$

$$\operatorname{Im} \left(\int_a^b w(t) dt \right) = \int_a^b \operatorname{Im}(w(t)) dt$$

Example:

$$\begin{aligned} \int_0^1 (1+it)^2 dt &= \int_0^1 1-t^2 dt + i \int_0^1 2t dt \\ &= \left. t - \frac{1}{3}t^3 \right|_0^1 + i \left. t^2 \right|_0^1 \\ &= \frac{2}{3} + i. \end{aligned}$$

We will not spend time describing when the integrals exist, as this is an involved discussion and better left to a class on real analysis. For us, it will be enough to note that by Riemann integration, the integrals exist when u, v are piecewise continuous with at most finitely many discontinuities.

Lemma: The following properties hold:

(1)
$$\int_a^b w(t) dt = \int_a^c w(t) dt + \int_c^b w(t) dt$$

(2) If $w(t) = u(t) + iv(t)$ and $W(t) = U(t) + iV(t)$ are defined for $t \in [a, b]$ and have $W'(t) = w(t)$ then

$$\begin{aligned} \int_a^b w(t) dt &= \int_a^b u(t) dt + i \int_a^b v(t) dt \\ &= U(b) - U(a) + i(V(b) - V(a)) = W(b) - W(a). \end{aligned}$$

Example: If $w(t) = e^{it}$ then as $\frac{d}{dt} \left(\frac{e^{it}}{i} \right) = e^{it}$ we expect that $W(t) = \frac{e^{it}}{i}$.

Indeed,

$$\begin{aligned} \int_0^{\pi/4} e^{it} dt &= \int_0^{\pi/4} \cos t dt + i \int_0^{\pi/4} \sin t dt \\ &= \sin t - i \cos t \Big|_0^{\pi/4} \\ &= \frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} - (-i1) \\ &= \frac{1}{i} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) - \frac{1}{i} \\ &= \frac{e^{i\pi/4}}{i} - \frac{e^{i0}}{i}. \end{aligned}$$

This fundamental theorem of calculus result suggests that no matter what path you take from $w(a)$ to $w(b)$, the result of the integral is the same! This is almost true, and it leads to the idea of a contour integral.

Contours in $\mathbb{C}$

To define integrals of complex-variable functions, we leverage properties of special paths.

Def: A set of points $z = (x, y) \in \mathbb{C}$ is an arc if

$$x = x(t) \text{ and } y = y(t) \text{ for } t \in [a, b] \subseteq \mathbb{R}$$

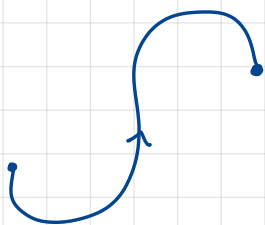
where x, y are continuous functions of t .

We will order the points z by their corresponding t values.

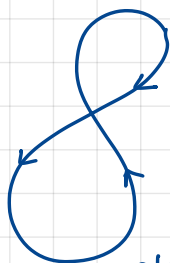
We also say that z is a simple arc or Jordan arc if the path is not self-intersecting.

If z is self-intersecting except for the endpoints i.e. $z(a) = z(b)$ we say z is a simple closed curve or Jordan curve. Any such curve is positively oriented when it is in the counterclockwise direction.

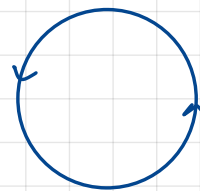
Examples:



simple arc

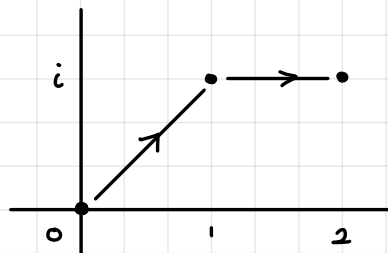


closed curve



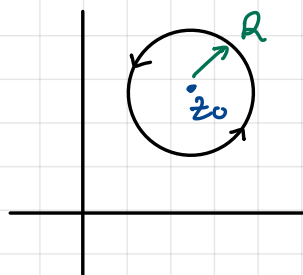
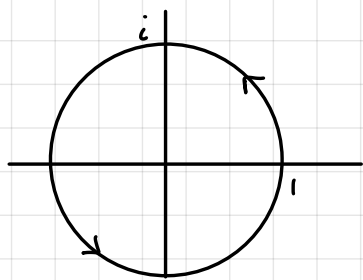
simple closed curve

Example:



$$z = \begin{cases} (1+i)t & 0 \leq t \leq 1 \\ t+i & 1 \leq t \leq 2 \end{cases}$$

simple arc



$$z = e^{it}, \quad 0 \leq t \leq 2\pi$$

$$z = z_0 + Re^{it}, \quad 0 \leq t \leq 2\pi$$

simple closed curves

The parametric representation of a path is not unique.
We can always use a change of variables

$$t = \phi(\tau) \quad \alpha \leq \tau \leq \beta, \quad \phi(\alpha) = a, \quad \phi(\beta) = b.$$

Assuming ϕ is continuous and differentiable, we assume $\phi'(\tau) > 0$ so that it is an increasing function.

Then
$$z(t) = z(\phi(\tau)) = Z(\tau)$$

If we differentiate z w.r.t. t we get

$$z'(t) = x'(t) + iy'(t).$$

The arc z is differentiable if it has a well-defined length

$$L = \int_a^b |z'(t)| dt = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt$$

Note that the length is invariant under parameterization

$$\begin{aligned}\int_{\alpha}^{\beta} \left| \frac{d}{d\tau} (z(\phi(\tau))) \right| d\tau &= \int_{\alpha}^{\beta} |z'(\phi(\tau))| |\phi'(\tau)| d\tau \\ &= \int_a^b |z'(t)| dt = L.\end{aligned}$$

Finally, if we assume $z'(t) \neq 0$ for all $t \in (a, b)$, then we can always define the unit tangent vector

$$\underline{T} = \frac{z'(t)}{|z'(t)|}$$

and this is a continuous function of t . Note the argument of $\underline{T}$ is $\arg(z'(t))$ and this is also continuous.

For this reason we call an arc $z = z(t)$ smooth if $z(t)$ is continuous for $t \in [a, b]$, differentiable for $t \in (a, b)$ and $z'(t) \neq 0$.

A contour C is a piecewise smooth arc
ie. it is made up of finitely many smooth arcs
joined end to end.

A simple closed contour C is a piecewise smooth closed curve.

Our goal is to determine what it means to find

$$\int_C f(z) dz.$$